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California Institute of Technology

Ph.D. in Computer Science
Advisor: Georgia Gkioxari

California Institute of Technology

2022 – 2026

B.S. in Computer Science

GPA: 4.3/4.3

Aligning Text, Images, and 3D Structure Token-by-Token

V. Tibrewal*, A. Sahoo*, G. Gkioxari

Computer Vision and Pattern Recognition Conference (CVPR) 2026.

Focus of Attention in Self-Supervised Learning for Action Recognition

V. Tibrewal, B. Thomson, M. Hugelshofer, H. Richter, P. Perona, N. Kondapaneni*, M. Marks*

CV4Smalls Workshop at CVPR 2026 [Best Paper Candidate].

CV4Animals Workshop at CVPR 2025.

Using Machine Learning over Fast Camera Images to Classify the Plasma Confinement Mode within the MAST Experiment

D. Brennand, V. Tibrewal, V. Gopakumar, S. Pamela, Z. Li, A. Anandkumar, The MAST team

49th EPS Conference on Plasma Physics 2023.

RESEARCH

Caltech, Georgia Gkioxari Lab

March 2025 – Present.

EXPERIENCE

Advised by Prof. Georgia Gkioxari

- Trained 3D tokenizer that encodes complex 3D shapes with an efficient token budget (512 tokens) for high-fidelity reconstruction and effective LLM decoding.
- Proposed a unified LLM framework aligning language, images and structured 3D scenes, a novel modality that encodes object-wise (shape, position) information.
- Trained 3D Multimodal LLM with unified next-token prediction to handle single-view 3D scene reconstructions from images, rendering 3D scenes to images, question-answering, instruction-following and real-world 3D object recognition.
- This co-first-author paper is accepted to CVPR 2026.

Caltech, Perona Vision Lab

January 2024 – February 2025

Advised by Prof. Pietro Perona

- Developed a method for improved self-supervised training of DINOv2 for animal action recognition, using SAM2’s zero-shot segmentation capabilities to focus DINOv2 on animals during training to overcome the object-centric bias of DINOv2.
- Applied improved DINOv2 feature extraction to identify statistically significant behavioral differences in groups of sheep post-neurosurgery from videos.
- Invited talk (for best paper candidate) at CV4Smalls Workshop at CVPR 2026.

Caltech, Anima AI + Science Lab

November 2022 – December 2023

Advised by Prof. Anima Anandkumar

- Researched fourier neural operators to model the magnetohydrodynamic equations governing plasma evolution in Tokamak nuclear fusion reactors.
- Trained binary classifier to classify the plasma confinement mode of a Tokamak nuclear fusion reactor from fast camera images from the MAST experiment.
- Presented as second-author work at 49th EPS Conference on Plasma Physics 2023.

WORK

Citadel, Core Data Team

June 2025 – August 2025

EXPERIENCE

Research Engineer Intern

- Designed and implemented multiple projects from scratch that are (1) live in production (2) handling real-time financial data at scale (3) and integrated into the research platform that serves all of Citadel's Global Quantitative Strategies.

HONORS AND AWARDS	Henry Ford II Scholar	2025
	For engineering students with the best academic record at the end of junior year.	
	Doris S. Perpall SURF Speaking Competition Semi-Finalist	2024
	For excellence in oral presentation at SURF Seminar Talks.	
	Summer Undergraduate Research Fellowship (SURF) Award	2024
	For summer research with Professor Pietro Perona.	
TEACHING	William H. and Helen Lang SURF Endowment Award	2023
	For summer research with Professor Anima Anandkumar.	
	Alexander P. and Adelaide F. Hixon Prize for Writing	2023
	For the best composition in a first-year course.	
	EE/CS 148B: Large Language and Vision Models (TA)	Spring 2026
	CS 24: Computing Systems (Head TA)	Fall 2024, 2025
TALKS	CS 2: Data Structures (Head TA)	Winter 2024, 2025
	CS 3: Software Design in C (TA)	Spring 2024, 2025
	CS 1x: Intermediate Computer Programming (TA)	Fall 2023
	Aligning Text, Images, and 3D Structure Token-by-Token	
	<u>Vansh Tibrewal</u> , Aadarsh Sahoo, Georgia Gkioxari	
	<i>Caltech Information Science and Technology Council Meeting 2025.</i>	
ACTIVITIES	Focus of Attention in Self-Supervised Learning for Action Recognition	
	<u>Vansh Tibrewal</u> , Neehar Kondapaneni, Markus Marks, Pietro Perona	
	<i>SURF Seminar at Caltech 2024.</i>	
	Forecasting Disruptions and L-H Transitions in Tokamak Nuclear Fusion Reactors Using Fourier Neural Operators and Convolutional Neural Networks	
	<u>Vansh Tibrewal</u> , Zongyi Li, Anima Anandkumar	
	<i>SURF Seminar at Caltech 2023.</i>	
RELEVANT COURSEWORK	Caltech Peer Advocate	2024 - Present
	Caltech Board of Control Representative	2023 - Present
	Caltech Hackathon Director of Logistics	2023 - 2024
	Caltech House Social Events Team	2023 - 2024
SKILLS	Advanced Topics in Vision, Advanced Topics in Machine Learning, GPU Programming, Probability & Statistics (Intro, Graduate Level), Linear Algebra (Intro, Graduate Level), Learning Systems, Machine Learning, Compilers, Systems, Algorithms, Networks.	
	Python, C, Java, JavaScript, C++, R, PyTorch, TensorFlow, Keras, CUDA, Git	